

PROJECT DESIGN PHASE-II

Technology Stack (Architecture & Stack)

Date	13 July 2026	Team ID	
Project Name	India's Agricultural Crop Production Analysis using Tableau		
Maximum Marks	4 Marks		

Technical Architecture:

The deliverable outlines the high-level data pipeline and architectural layout mapping the flow from the raw agricultural data store to the public interactive visualization dashboard view:

[Agricultural Datasets] → [Microsoft Excel] → [Tableau Desktop / Public] → [Tableau
Public Server] → [Web Browser]
(Raw Data Source) (Data Cleansing & (Interactive Visual Analytics
(Cloud Publishing & (End-User UI
.csv / .xlsx Transformation) & Dashboard Engineering)
Hosting) Consumption)

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Provides an interactive graphical environment for end-users to filter, drill down, and explore dynamic crop production maps, trend lines, and regional KPIs.	Tableau Interactive Dashboard View / Modern Web Browser (Chrome, Firefox, Safari, Edge)
2	Application Logic-1	Handles the data model execution, calculated field computations, geographical role mapping, aggregation queries, and rendering parameters.	Tableau VizQL Engine (Tableau Desktop / Tableau Public)
3	Application Logic-2	Facilitates structural sheet relationships, extract filters, logical layer joining, and worksheet-to-dashboard action paths.	Tableau Desktop Worksheet & Story Workspace
4	Database	Structured, non-relational flat-file tabular data storing crop parameters (State, District, Crop Year, Season, Crop, Area, Production).	Microsoft Excel Worksheet Data Store (.xlsx / .csv)
5	Cloud Database		Tableau Public Cloud Repository

S.No	Component	Description	Technology
		Managed analytics hosting environment that serves compiled dashboard workbooks and houses extract files for public web delivery.	
6	File Storage	Local system filesystem utilized for staging source worksheets and persisting packaged workbook files before publishing.	Local Filesystem (.twbx / .xlsx)
7	Infrastructure (Server / Cloud)	Secure, scalable cloud infrastructure that hosts the final interactive agricultural workbook for public availability and access.	Tableau Public Server Web Architecture

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Utilizes core built-in visualization and computation libraries embedded within the analytical platform to manage vector mapping and chart renders.	Tableau Presentation & Layout Components
2	Security Implementations	Employs encrypted visualization transport over the web, secure extraction layers, and robust multi-tenant authorization controls managed by the hosting repository.	HTTPS / SSL Encryption, Tableau Public Platform Access Controls
3	Scalable Architecture	Relies on an efficient client-server layout where heavy client-side analytics are decoupled from high-throughput web servers serving global viewer sessions.	Tableau Cloud Multi-tenant Web Tier Architecture
4	Availability	Guarantees high global availability through robust server caching networks, distributed database shards, and managed platform redundancy.	Tableau Public Content Delivery Networks (CDNs)
5	Performance	Optimized performance achieved via Hyper-extracted tabular compilation, analytical data aggregation layers, and client-side browser caching mechanisms.	Tableau VizQL Query Caching & Hyper Data Engine Technology

References:

- <https://c4model.com/>
- <https://help.tableau.com/current/pro/desktop/en-us/default.htm>
- <https://www.ibm.com/cloud/architecture>